Day 069 | Unit 05 | Teacher plan

Represent and Solve Equations and Inequalities

Test Inequality Solutions

Standards: 6.EE.B.5, 6.EE.B.8

Learning target: I can test values and explain the many solutions of an inequality.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Substitute each value and decide whether the comparison is true. A continuous number-line ray represents infinitely many real-number solutions. A real situation may restrict useful solutions to whole numbers.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 3, 25

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01 / Learn

Substitute each value and decide whether the comparison is true. A continuous number-line ray represents infinitely many real-number solutions. A real situation may restrict useful solutions to whole numbers.

Worked example 1

Which values in {5, 6, 7} satisfy $x \le 6$? Test each by substitution.

1. Substitute each value and decide whether the comparison is true.

2. A continuous number-line ray represents infinitely many real-number solutions.

3. Values below 6 work

4. the endpoint works too.

5. State the result in the form requested: 5, 6.

Answer: 5, 6

Worked example 2

A club needs at least 18 members. Select all true statements for its membership count n.
A. $n \ge 18$
B. $n > 18$
C. 17 is an allowed membership count.
D. 18 is an allowed membership count.

1. Substitute each value and decide whether the comparison is true.

2. At least includes the boundary, so $n \ge 18$.

3. On a real number line use a closed point and shade right

4. actual member counts are whole numbers.

Answer: A, D - $n \ge 18$; 18 is an allowed membership count.

Day 069 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Which values in {6, 7, 8} satisfy x < 7? Test each by substitution.

02 / Constructed response

Which values in {11, 12, 13} satisfy x ≤ 12? Test each by substitution.

03 / Constructed response

Which values in {14, 15, 16} satisfy x < 15? Test each by substitution.

04 / Constructed response

Which values in {10, 11, 12} satisfy x ≤ 11? Test each by substitution.

05 / Constructed response

Which values in {5, 6, 7} satisfy x < 6? Test each by substitution.

06 / Constructed response

Which values in {8, 9, 10} satisfy x ≤ 9? Test each by substitution.

07 / Constructed response

Which values in {8, 9, 10} satisfy $x < 9$? Does the real-number inequality have only these solutions? Explain.

08 / Multiple select

A club needs at least 6 members. Select all true statements for its membership count n.
A. $n > 6$
B. 5 is an allowed membership count.
C. $n \geq 6$
D. 6 is an allowed membership count.

09 / Multiple select

A club needs at least 10 members. Select all true statements for its membership count n.
A. $n > 10$
B. 10 is an allowed membership count.
C. 9 is an allowed membership count.
D. $n \geq 10$

10 / Constructed response

Which values in {10, 11, 12} satisfy $x \leq 11$? Does the real-number inequality have only these solutions? Explain.

Day 069 | Unit 05 | Answer key

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1. 6

Values below 7 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

2. 11, 12

Values below 12 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

3. 14

Values below 15 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

4. 10, 11

Values below 11 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

5. 5

Values below 6 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

6. 8, 9

Values below 9 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

7. 8; no, infinitely many real values satisfy it

Values below 9 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

8. C, D - n ≥ 6; 6 is an allowed membership count.

At least includes the boundary, so n ≥ 6. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.5, 6.EE.B.8 | Multiple select

9. B, D - 10 is an allowed membership count.; n ≥ 10

At least includes the boundary, so n ≥ 10. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.5, 6.EE.B.8 | Multiple select

10. 10, 11; no, infinitely many real values satisfy it

Values below 11 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response